

ABDULLAH AL JUBAER GEM

jubaer@iut-dhaka.edu | linkedin.com/in/abdullah-al-jubaer | github.com/jubaer | +8801736587392 | kaggle.com/jubaer36

ABOUT ME

Almost everything in life piques my interest, whether it be simple everyday curiosities or complex mathematical problems whose solutions are yet to be discovered. Accepting that I could not pursue every interest, I chose Computer Science as my primary field of exploration, where I found myself drawn to the intersection of machine learning and full-stack development. Since then, I have pursued that interest through research, technical projects, and competitive datathons. I currently lead the software team for Project Aqua, an autonomous ROV, while conducting my undergraduate thesis on Multimodal Emotion Recognition in Conversations (MERC), continually expanding my expertise in machine learning, computer vision, and software engineering.

EDUCATION

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)-OIC	2022 – September 2026 (Expected)
<i>BSc. in Computer Science And Engineering</i>	<i>CGPA: 3.75</i>
PABNA CADET COLLEGE	2021
<i>Higher Secondary Certificate (HSC)</i>	<i>GPA: 5.00</i>
PABNA CADET COLLEGE	2019
<i>Secondary School Certificate (SSC)</i>	<i>GPA: 5.00</i>

SKILLS

Languages	Python, Java, JavaScript, TypeScript, HTML5, CSS3
Frameworks	PyTorch, TensorFlow, Keras, React.js, Next.js, Express.js, Spring Boot
Libraries	NumPy, scikit-learn, Pandas, Tailwind CSS, shadcn/ui
Databases	PostgreSQL, MongoDB, MySQL, Firebase
Tools & Platforms	Git, GitHub, Linux (Ubuntu, Arch), Docker, ROS2, Arduino, Pixhawk, Blender
Relevant Courses	Object Oriented Programming, Machine Learning, Data Structures, Algorithms, Software Engineering and OOD, Database Management System, Structured Programming, Linear Algebra, Web Architecture, Operating Systems, Probability and Statistics, Computer Graphics and Multimedia Systems, Digital Image Processing, Visual Programming, Numerical Methods, Theory Of Computing, Artificial Intelligence, Compiler Design, Pattern Recognition, Human Computer Interaction, Graph Theory

WORK EXPERIENCE

Medi Merchandise (Startup)	Remote
<i>Full Stack Developer</i>	
- Built a platform managing supply chain, sales, and investment for a medical merchandise startup. - Designed multi-role portals (Admin, Merchant, Investor) with inventory, procurement, and automated ROI.	
Project Aqua	University Club
<i>Software Co-Lead</i>	
- Leading the software team building control and automation for a remotely operated vehicle (ROV). - Prototyped a semi-autonomous drain-cleaning bot.	
Alikbachan– Bengali LLM Hallucination Detection Challenge	IUT Datathon
<i>Event Head</i>	
- Event Head for IUT's datathon, running the full pipeline from dataset to judging for 250+ teams. - Led dataset curation and problem design for a Bengali LLM hallucination detection task.	

RESEARCH EXPERIENCE

Graph Based Multimodal Emotion Recognition in Conversations	Undergraduate Thesis, In Progress
- Conducted module level ablations across SOTA and isolated which components drive performance versus which are redundant, informing design choices for confusable emotion class discrimination. - Designed multiple graph based fusion architectures to address modality imbalance, and underutilization of the visual modality and similar emotion confusion problem.	
Alikbachan: Bengali LLM Hallucination Benchmark	Submitted to ACL, Under Review

- Tested frontier AI models on a 7,500+ item Bengali benchmark to see how well they hallucinated answers.
- Generated model confidence scores (log-probabilities) on benchmark responses to test if they could flag token level hallucinations.

EthMuSAM: Zero-Shot Industrial Anomaly Detection

CVPR Workshop

- Zero-shot framework that detects defects in unseen product types without training.
- DINOv3-MuSc scoring locates suspect regions; cascaded SAM refines precise defect masks.

Multimodal Image Fusion: Medical CT-MRI Image Fusion

Machine Learning Course Project

- Fused CT and MRI scans into one image, keeping complementary detail for diagnosis.
- Curated a paired CT-MRI dataset; compared deep-learning and wavelet fusion methods.

Weakly Supervised Named Entity Recognition via Sparse-CHMM and BERT

Mathematical Analysis Course Project

- Aggregates noisy labeling functions without ground-truth annotations.
- Models sequence dependencies using Sparse-CHMM with contextual embeddings.

PROJECTS

BizTrack – AI-Driven Business Management Platform | *Next.js, Express.js, Supabase, TypeScript, Docker*

- Full-stack platform with AI-driven financial analysis, risk alerts, and weather-aware demand forecasting.
- Next.js dashboard with customer/retention analytics; Supabase auth, Dockerized deployment.

Redrick-Routinson: University Management Webapp | *Python, Reactjs, Chakra UI, Flask, Firebase*

- Role-based university platform for admins, teachers, and students.
- Genetic-algorithm class scheduling, automated exam seat planning, and QR attendance.

No-Brainer: AI-Powered Fitness Program Generator | *Next.js, Reactjs, TailwindCSS, Convex, Clerk, Google Generative AI, VAPI*

- AI-generated workout and diet plans with voice-assistant onboarding (VAPI, Google Generative AI).
- Clerk auth, real-time Convex backend, deployed on Vercel.

Student Marketplace | *Reactjs, Firebase, Vite, GROQ API*

- Student trading platform with real-time chat and bidding.
- AI item analysis and chatbot via GROQ API; Firebase auth and admin controls.

Ruby: Learn the Rubik's Cube | *Java, JavaFX, Swing*

- Desktop app for stepwise Rubik's Cube solving with interactive visualizations.
- Manual cube-state input and automated solution demo; built in JavaFX/Swing.

The Continental: Hotel Management System | *React, Node.js, MySQL*

- Hotel management with room booking, check-in/out, and admin controls.
- React/Node.js/MySQL stack with JWT authentication.

AWARDS, ACHIEVEMENTS & SCHOLARSHIPS

5th Place – VAND 4.0 Industrial Track

CVPR 2026 Challenge

Zero-shot, class-agnostic industrial anomaly detection on unseen categories. **[Leaderboard]**

Runners-Up – Coderush 2025

IUT Hackathon

24 hour hackathon to build a working full stack product from scratch on a surprise theme.

7th Place – BUBT Datathon 2025

24-Hour National Datathon

National datathon to build predictive models on a real world tabular dataset.

Participant – SP Cup

IEEE Signal Processing Cup

Deepfake detection with fine-tuned ML models; wrote a technical report. **[Report]**

Partial OIC Scholarship – Awarded by the Islamic University of Technology

2022–2026

Gold Medal – Shahid Cadet Coaching

2015

Ranked 3rd out of 1500+ students in the final exam.

Board Scholarships – Merit scholarships in PSC, SSC, and HSC examinations.

SOCIAL ENGAGEMENTS

University Clubs: Project Aqua , Project Aero , IUTCS

Hobbies: Reading, Watching TV Series and Movies , Writing

Sports: Table Tennis (Captain, IUT), Basketball (Vice-Captain, CSE), Badminton

REFERENCES

- **Shahriar Ivan** – Assistant Professor, Department of CSE, IUT
Specialization: Machine Learning, Deep Learning, Computer Vision
shahriarivan@iut-dhaka.edu